

Marketing Campaign Analysis — Phase 6: Market Basket Analysis

Omar Touzani · Apriori + FP-Growth on 2,212 customers · 6 product categories · per-CLV-tier breakdown

The spending data is 2-year aggregate per customer per category — not transactions. To run association-rule mining meaningfully, each customer is treated as an "above-median buyer" of a category if they spent above the median for that category. The output identifies **purchasing affinities** — which categories tend to attract the same customers — which is what's actionable for cross-sell.

Headline findings

Fish is the connector category. Across every CLV tier and the catalog-heavy segment, Fish appears in nearly every top rule. Customers who buy above-median in Fish overwhelmingly also buy above-median in Fruits, Sweets, and Meat. Fish is the strongest single basket-anchor in the catalogue.

Catalog-heavy customers are a fresh-food affinity segment. The strongest rule among catalog-heavy customers is Fish + Fruits ↔ Meat + Sweets (lift 1.69). Wines and Gold don't appear in their top rules. This connects directly to the Phase 3 finding that catalog is the luxury channel — those luxury customers are skewing toward fresh-food bundles, not wine + premium combinations.

Basket affinity weakens as CLV rises — the opposite of what cross-sell intuition predicts. The Low value tier shows the strongest category preferences (lifts up to 1.85); the High value tier shows the weakest (lifts max out at 1.42), with zero rules unique to it. Implication: high spenders buy above-median in every category, so they have no distinguishing "basket signature" — "you bought X, try Y" cross-sell tactics are **less** effective on top customers, not more. They already buy everything.

Per-CLV-tier comparison

CLV tier	Customers	Rules found	Strongest lift	Interpretation
Low value	738	55	1.85	Sharp category preferences — Fruits ↔ Fish+Gold
Mid value	737	108	1.87	Most rules — clearest cross-sell signal
High value	737	25	1.42	Buy across all categories — no distinguishing basket

Top rules by lift — overall (pair and triple rules only)

Antecedent	Consequent	Support	Confidence	Lift
Fruits, Gold	Fish	26.0%	73.4%	1.80
Gold, Sweets	Fish	26.0%	72.5%	1.71
Fish, Sweets	Fruits	27.8%	69.4%	1.70
Fish, Fruits	Sweets	27.8%	68.7%	1.62
Sweets	Fish	31.3%	63.2%	1.56
Fish	Fruits	31.4%	64.0%	1.64

Recommendations

For cross-sell campaigns, target Mid value customers first — they show the strongest, most distinctive category affinities and 108 actionable rules. Lead with Fish-anchored bundles (Fish + Fruits, Fish + Sweets, Fish + Gold + Fruits).

For catalog-channel campaigns, design a fresh-food bundle (Fish + Fruits + Meat + Sweets). This pattern is specific to catalog-heavy buyers and aligns with the catalog channel's role as the high-value engagement surface identified in Phase 3.

For High value customers, avoid product-affinity cross-sell entirely. Their basket shows no signature preferences — they buy across the board. Marketing to them should focus on volume incentives, exclusivity, or loyalty mechanics rather than "you bought X, try Y."

Methodology caveat

The above-median binary signal is a defensible adaptation of association-rule mining to aggregate spend data, but it's not a substitute for true transaction-level basket analysis. The rules identify *customer affinities across categories* over a 2-year window, not literal items appearing in single shopping trips. If transaction logs become available in future, this analysis should be re-run on real basket data.

What's next

Tableau dashboard build. Four pages: Campaign Performance (Phase 3), CLV & Targeting (Phase 4), Sentiment (Phase 5), Market Basket (Phase 6). All processed CSVs from Phases 4–6 are saved in *data/processed/* ready for Tableau ingestion.